

Marcos A. Cruz Bazan

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PROFESSIONAL SUMMARY

Computer Science student with hands-on experience building full-stack applications, backend APIs, low-level networking systems, and explainable machine learning risk engines. Proficient in Python, C++, TypeScript, SQL, and Flutter, with projects spanning FastAPI services, PostgreSQL and Azure SQL data systems, containerized deployments, and real-time security telemetry.

EDUCATION

Florida International University

Bachelor of Science in Computer Science.

Miami, FL

Expected December 2026

Miami Dade College

Associate of Arts in Computer Science. Presidential Scholars Scholarship Recipient; Member of Phi Theta Kappa Honor Society.

December 2023

EXPERIENCE

Student Office Associate, Scanning Operations - COB Graduate Enrollment

October 2024 – Present

Florida International University

Miami, FL

- Execute document processing requests submitted by program recruiters within Salesforce CRM, improving data accuracy across student files.
- Track and manage application life-cycle updates, including test score record matching, program metrics, term transitions, and operational reporting.
- Serve as a communication bridge between technical operations, recruiters, and academic administrators while maintaining compliance with privacy regulations.

PROJECTS

StockFW - Intelligent Warehouse Operating System

FastAPI, PostgreSQL 16, Flutter, Scikit-Learn, Docker Compose, Caddy, Alembic

- Built a containerized full-stack warehouse platform with FastAPI, PostgreSQL, Flutter, Caddy TLS/CORS handling, OAuth2/JWT security, scheduled PostgreSQL backups, and GitHub Actions CI/CD.
- Implemented an ACID-compliant inventory ledger with Alembic migrations and atomic stock deductions and credits for bin-to-bin transfers.
- Integrated Economic Order Quantity (EOQ) replenishment with an Isolation Forest model that flags abnormal movement patterns for shrinkage and theft review.

TruthScore - Access-Risk Scoring & Explainable AI Platform

TypeScript, Node.js, Hono, Flutter, Python, FastAPI, Azure SQL / SQL Server, Scikit-Learn, Drizzle ORM

- Designed context-aware 0-100 access-risk scoring using device trust, IP/location, access hour, behavioral baselines, and peer norms, returning Allow, MFA, Review, or Block decisions with plain-English explanations.
- Built a Hono/Node API, Flutter client, Azure SQL/SQL Server data layer with parameterized T-SQL, and Python FastAPI ML services using Random Forest and Isolation Forest models.
- Added append-only adjudication audit trails, multi-tenant client scoping, emergency lockdown, graceful degradation, OpenTelemetry tracing, Docker Compose orchestration, and GitHub Actions CI/CD.

EntropyShield - Real-Time Data Leak Prevention Gateway

Python, FastAPI, Shannon Entropy, WebSockets, Luhn Validation

- Engineered a FastAPI JSON gateway that inspects outbound payloads before forwarding them, applying ALLOW, MASK, and BLOCK policy decisions to reduce runtime leakage of sensitive data.
- Combined pattern detectors for AWS/Stripe keys, email, SSN, JWTs, and Luhn-valid card numbers with Shannon entropy scoring for high-randomness secrets.
- Decoupled live dashboard telemetry from request handling with a queue and WebSocket event stream so slow monitoring clients do not block gateway traffic.

SchemaFlow - Visual SQL Schema Designer

Flutter Web, Dart, CustomPainter, SQL DDL

- Built an interactive visual SQL schema designer with a custom vector engine, infinite canvas, draggable table nodes, and real-time relationship connectors.
- Implemented a live transpilation pipeline that maps relational state to formatted ANSI-SQL DDL using native Flutter/Dart APIs and a decoupled, data-driven state architecture.

Socket FTP Protocol Client

Python, Raw BSD Sockets, TCP/IP, RFC 959

- Implemented an RFC 959 FTP client directly over raw BSD TCP sockets without third-party networking abstractions or wrappers.
- Handled stateful TCP control handshakes, binary stream transfers, and raw packet byte parsing for file transport.
- Implemented dynamic PASV port decoding using bit shifting and arithmetic recombination ($P1 * 256 + P2$) to establish parallel data channels.

SKILLS

Frameworks & Libraries: FastAPI, Flutter, Node.js, Hono, Scikit-Learn, SQLAlchemy ORM, Drizzle ORM

Programming Languages: Python (AsyncIO, Raw Sockets), TypeScript, JavaScript, SQL, C++, Dart, Swift, Java, C

Databases & Data: PostgreSQL, Microsoft SQL Server / Azure SQL, SQLite, T-SQL, Alembic, Relational Modeling

DevOps, Systems & Security: Docker, Docker Compose, Caddy, Git, GitHub Actions, Linux, WebSockets, Raw TCP/IP Sockets, Zero-Trust IAM, OpenTelemetry

CERTIFICATIONS

- App Development with Swift - Apple (Issued: May 2024)